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1 Name of Experiment

Manual Black-Box Testing and Bug Discovery

on a Deliberately Buggy E-Commerce Web Application

Application: `https://with-bugs.practicesoftwaretesting.com` | Toolshop v5.0 (Buggy Version)

This experiment involves performing structured manual testing across all major functional modules of the **Toolshop** e-commerce web application. The application is a publicly available practice platform intentionally seeded with defects to simulate real-world software quality challenges. Testers are required to discover, reproduce, document, classify, and report every defect following an industry-standard bug-reporting format.

2 Objective

he objectives of this laboratory experiment are as follows:

1. To perform comprehensive **manual black-box testing** on a multi-module e-commerce web application from the perspective of an end-user, without access to the source code.

2. To **identify, reproduce, and document software defects** across all functional areas including navigation, authentication, product listing, shopping cart, checkout, user account management, and the admin panel.

3. To **classify each discovered bug** by severity level (Critical, High, Medium, Low) and provide a fully structured reproduction path with test data.

4. To practise industry-standard **defect reporting** using a formal bug tracking template covering Bug ID, Module, Related Requirement, Test Case ID, Steps to Reproduce, Expected vs. Actual Result, Severity, and Status.

5. To develop analytical skills in **identifying security vulnerabilities**, including Cross-Site Scripting (XSS), broken access control, and missing input sanitisation.

6. To gain hands-on experience applying **exploratory and scenario-based testing techniques** on a realistic web application environment.

3 Theory

3.1 Software Testing

Software testing is the systematic process of evaluating and verifying that a software product behaves as intended. Its primary goal is to identify errors, gaps, or deviations from requirements before the product reaches end users. Testing can be broadly divided into **manual testing** (performed by human testers) and **automated testing** (executed by scripts and tools).

3.2 Black-Box Testing

Black-box testing evaluates the external behaviour of an application without any knowledge of its internal code structure. The tester only knows *what the system should do*, not *how it does it*. Inputs are provided and outputs are observed against expected results. This technique is particularly effective for functional, integration, and user acceptance testing. All test cases in this experiment were designed and executed using the black-box approach.

3.3 Types of Defects Encountered

- **Functional Defects** — A software feature does not behave as specified (e.g., a category filter that shows all products, a sort that has no effect).
- **Input Validation Defects** — Missing or incorrect checks that allow invalid, empty, or malicious data to be accepted by the system (e.g., blank form submission, weak passwords).
- **UI / Display Defects** — Visual components that render incorrectly, such as broken images, missing counters, or unresponsive interface elements.
- **Security Defects** — Vulnerabilities that can be exploited by malicious users, including Cross-Site Scripting (XSS) and broken role-based access control.
- **Data Integrity Defects** — Errors in calculations, state not persisted to the backend, or data discarded silently (e.g., wrong cart total, profile changes not saving).
- **Workflow / Navigation Defects** — Missing redirects, incorrect page routing, or absent authentication guards on protected routes.

3.4 Bug Severity Classification

Severity	Definition
Critical	Causes complete system failure, data loss, financial error, or a critical security vulnerability. Must be resolved before any release.
High	Significantly impairs a core feature with no viable workaround. Blocks further testing in that module.
Medium	Partially impairs functionality; a workaround may exist. Should be resolved before the next release cycle.
Low	Minor cosmetic or low-impact usability issue with no effect on core functionality.

3.5 Application Under Test: Toolshop v5.0 (Buggy Version)

The **Toolshop** is a simulated e-commerce web application built with an Angular front-end and a RESTful JSON backend API. It sells hardware and tools across several categories including Hand Tools, Power Tools, Plumbing, and Miscellaneous. The *with-bugs* version at `https://with-bugs.practicesoftwaretesting.com` contains intentional defects seeded across its modules to support QA training and practice.

The modules covered in this experiment are:

- Navigation & Header
- User Registration & Login
- Product Listing, Search & Filtering
- Product Detail Page
- Admin Panel
- Shopping Cart
- Checkout & Payment
- User Account (Profile, Orders, Password)
- Contact Form

4 Test Cases

A total of **25 test cases** were designed and executed across all major functional modules of the Toolshop e-commerce application, including Navigation, Search, Product

Listing, Shopping Cart, User Authentication, Checkout, and Admin Panel.

Each test case was structured with a unique Test Case ID, associated Bug ID, module name, severity level, step-by-step execution procedure, test data, expected result, and actual observed result.

Due to space constraints, only a few representative test cases are presented below. These samples highlight critical, high, and medium severity defects identified during testing. The complete set of test cases, along with detailed results and screenshots, has been documented separately in the accompanying Excel report for better clarity and traceability.

Sample Test Cases:

TC-0018 | Bug ID: **BG-0009** | Shopping Cart — Total Calculation | **Critical**

Steps	1. Add Item 1 with quantity 2 to cart 2. Add Item 2 with quantity 1 to cart 3. Open the cart page
Test Data	Item 1: \$12.99 × 2 Item 2: \$24.99
Expected	Cart total should be calculated correctly based on quantity and price
Actual	**Incorrect total displayed; quantity multiplier is not applied properly**

TC-0012 | Bug ID: **BG-0003** | Navigation — Category Filter | **High**

Steps	1. Open Homepage 2. Select a category (Hand Tools) from dropdown
Test Data	Category: Hand Tools
Expected	Only products from the selected category should be displayed
Actual	**All products are displayed; category filter has no effect**

TC-0014 | Bug ID: **BG-0005** | Product Listing — Sort by Price | **Medium**

Steps	1. Open product listing page 2. Select "Price: Low to High" sorting option
Test Data	Sort option: Price Low to High
Expected	Products should be arranged from lowest to highest price
Actual	**Sorting has no effect; default order remains unchanged**

TC-0033 | Bug ID: **BG-0024** | Search — XSS Vulnerability | **Critical**

Steps	1. Open Homepage 2. Enter `<script>alert(1)</script>` in search bar 3. Submit search
Test Data	Malicious script input
Expected	Input should be sanitised and no script should execute
Actual	**JavaScript alert executes, confirming XSS vulnerability**

The above test cases demonstrate the presence of critical functional, validation, and security issues within the system. These defects significantly impact system reliability, user experience, and application security, and require immediate attention before deployment.

5 Screenshots

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Bug ID	Module	Related Requirement	Test Case ID	Reproduction Path	Test Data	Expected Result	Actual Result	Severity	Status	Test Environment	Test Case Author	Executed By	Bug Reported by	Attachment	Remarks
2	BG-0003	Navigation	Category Filtering	TC-0012	1. Open Homepage 2. Click 'Categories' dropdown 3. Select 'Hand Tools'	N/A	Page should display only Hand Tools products	Page displays all products regardless of selected category; filter has no effect	High	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Filter logic not applied on navigation category click
3	BG-0004	Search	Product Search	TC-0013	1. Open Homepage 2. Type 'Hammer' in the search bar 3. Press Enter	Search query: 'Hammer'	Search results should display all products matching 'Hammer'	Search returns zero results even though matching products exist in the catalogue	High	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Search index appears broken or case-sensitive mismatch
4	BG-0005	Product Listing	Sort Functionality	TC-0014	1. Open any category page 2. Select 'Price: Low to High' from sort dropdown	N/A	Products should be reordered from lowest to highest price	Products remain in default order; sort selection has no effect on displayed products	Medium	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Sort parameter not passed to API or UI not re-rendering
5	BG-0006	Product Details	Product Image Display	TC-0015	1. Open Homepage 2. Click on 'Thor Hammer' product 3. View product detail page	Product: Thor Hammer	Main product image should load and display correctly	Product image shows a broken image icon; image URL returns 404	Medium	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Broken image asset or wrong image path in product data
6	BG-0007	Shopping Cart	Add to Cart	TC-0016	1. Open any product detail page 2. Click 'Add to Cart' button	Product: Any in-stock item	Item should be added to the cart and cart counter in header should increment by 1	Item appears added but cart counter in header does not update; counter remains at 0	High	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Cart badge counter not reactive to state changes
7	BG-0008	Shopping Cart	Cart Quantity Validation	TC-0017	1. Open Cart 2. Change quantity of an item to 0 3. Click Update	Quantity: 0	System should show validation error or automatically remove item from cart	Item quantity is set to 0 but item remains in cart with a total of $0.00, causing incorrect order total	High	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	No minimum quantity validation on cart input
8	BG-0009	Shopping Cart	Cart Total Calculation	TC-0018	1. Add multiple items with different quantities to cart 2. View cart total	Item 1: $12.99 x2, Item 2: $24.99 x1	Cart total should be $50.97	Cart displays incorrect total ($37.98); one item's quantity multiplier not applied	Critical	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Price multiplication logic broken for multiple quantities
9	BG-0010	User Registration	Password Strength Validation	TC-0019	1. Navigate to /auth/register 2. Fill all fields 3. Enter password: '123' 4. Click Register	Password: '123'	System should reject weak password and display strength requirement message	Registration succeeds with a 3-character numeric-only password; no validation enforced	High	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Password policy not enforced on registration endpoint
10	BG-0011	User Registration	Duplicate Email Validation	TC-0020	1. Navigate to /auth/register 2. Enter existing email: customer@practicesoftwaretesting.com 3. Fill remaining fields 4. Click Register	Email: customer@practicesoftwaretesting.com	System should display error: 'An account with this email already exists'	Page reloads without any error message; user is left confused with no feedback	Medium	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Duplicate email error not surfaced to UI
11	BG-0012	User Login	Successful Login Redirect	TC-0021	1. Navigate to /auth/login 2. Enter email: customer@practicesoftwaretesting.com 3. Enter password: welcome01 4. Click Login	Email: customer@practicesoftwaretesting.com Password: welcome01	User should be redirected to the homepage and see their name in the header	Login succeeds but user is redirected to a blank page ('/'); no authenticated state visible in header	High	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Post-login redirect or token storage issue

Figure 1: Screenshot showing Test Cases TC-0001 to TC-0010 from the Excel Sheet

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Bug ID	Module	Related Requirement	Test Case ID	Reproduction Path	Test Data	Expected Result	Actual Result	Severity	Status	Test Environment	Test Case Author	Executed By	Bug Reported by	Attachment	Remarks
12	BG-0013	User Account	Profile Update	TC-0022	1. Login as customer 2. Navigate to Account > Profile 3. Change First Name 4. Click Save	New First Name: 'TestUser'	Profile should be updated and success message shown: 'Your profile has been saved'	Page shows success toast briefly but on refresh, the old name is displayed; changes not persisted	High	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	PATCH /users/{id} either not called or returns error silently
13	BG-0014	User Account	Change Password	TC-0023	1. Login as customer 2. Go to Account > Change Password 3. Fill all fields with valid new password 4. Click Update	Current: welcome01 New: NewPass@123 Confirm: NewPass@123	Password should be updated; user should remain logged in or be prompted to re-login	Form submits, shows no feedback (no success or error message), and password is not actually changed	High	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Change password API response not handled in UI
14	BG-0015	Contact Form	Required Field Validation	TC-0024	1. Navigate to Contact page 2. Leave all fields empty 3. Click Send	All fields: empty	Validation errors should highlight all required fields (Name, Email, Message)	Form submits successfully with empty fields; a success message appears with no data sent	High	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Frontend and backend validation both missing on contact form
15	BG-0016	Contact Form	Email Format Validation	TC-0025	1. Navigate to Contact page 2. Enter invalid email: 'notaremail' 3. Fill other required fields 4. Click Send	Email: 'notaremail' First Name: Test Last Name: User Message: Hello	System should display: 'Please enter a valid email address'	Form submits successfully with an invalid email format; no validation error shown	Medium	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Email regex validation not implemented on Contact form
16	BG-0017	Checkout	Checkout Without Login	TC-0026	1. Add item to cart (without logging in) 2. Go to cart 3. Click 'Proceed to Checkout'	User: Guest (not logged in)	User should be redirected to login page with message: 'Please log in to continue checkout'	User is taken to checkout page without authentication; order form is shown with no user data	Critical	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Auth guard missing on checkout route
17	BG-0018	Checkout	Payment Form Validation	TC-0027	1. Login and add item to cart 2. Proceed to checkout 3. Leave payment fields blank 4. Click Confirm	All payment fields: empty	Validation should prevent order submission and highlight required fields	Order is placed successfully with blank payment details; no validation triggered	Critical	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Payment form has no client-side or server-side validation
18	BG-0019	Checkout	Order Confirmation Email	TC-0028	1. Login and complete a full checkout with valid data 2. Confirm order	Email: customer@practicesoftwaretesting.com	User should receive order confirmation email after successful checkout	No confirmation email is received after order placement; no email trigger in system logs	Medium	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Email service not called on order creation
19	BG-0020	Product Details	Out-of-Stock Add to Cart	TC-0029	1. Find a product marked as 'Out of Stock' 2. View product detail page 3. Attempt to click Add to Cart	Product: Any out-of-stock item	'Add to Cart' button should be disabled or hidden for out-of-stock products	Out-of-stock product can still be added to cart; no stock validation enforced	High	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Stock status check not applied to Add to Cart action
20	BG-0021	Product Listing	Pagination	TC-0030	1. Open any category with more than 9 products 2. Click page '2' in pagination	N/A	Page 2 of products should load with the next set of items	Clicking page 2 returns the same products as page 1; page parameter ignored in API call	Medium	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Pagination offset/limit not passed correctly to backend
21	BG-0022	Product Details	Product Rating Submission	TC-0031	1. Login as customer 2. Open product detail page 3. Select star rating and write review 4. Click Submit	Rating: 4 stars Review: 'Good product'	Review should be saved and displayed below the product with author name	Review form submits but the review does not appear on the product page even after refresh	Medium	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Review POST succeeds but GET for reviews not returning new entry

Figure 2: Screenshot showing Test Cases TC-0011 to TC-0020 from the Excel Sheet

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Bug ID	Module	Related Requirement	Test Case ID	Reproduction Path	Test Data	Expected Result	Actual Result	Severity	Status	Test Environment	Test Case Author	Executed By	Bug Reported by	Attachment	Remarks
21	BG-0022	Product Details	Product Rating Submission	TC-0031	1. Login as customer 2. Open product detail page 3. Select star rating and write review 4. Click Submit	Rating: 4 stars Review: 'Good product'	Review should be saved and displayed below the product with author name	Review form submits but the review does not appear on the product page even after refresh	Medium	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Review POST succeeds but GET for reviews not returning new entry
22	BG-0023	User Account	Order History Display	TC-0032	1. Login as customer who has placed orders 2. Navigate to Account > My Orders	User: customer@practicesoftwaretesting.com	All past orders should be listed with order ID, date, total, and status	Order history page is blank/empty even for users with confirmed past orders	High	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Orders endpoint returning empty or user ID mismatch in query
23	BG-0024	Product Search	Search With Special Characters	TC-0033	1. Open Homepage 2. Enter special characters in search bar: '<script>alert()</script>' 3. Press Enter	Search query: '<script>alert()</script>'	System should sanitize input and show: 'No results found' without executing any script	Alert dialog executes in the browser; XSS vulnerability confirmed — script is not sanitized	Critical	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	XSS vulnerability — user input not sanitized before rendering in DOM
24	BG-0025	Navigation	Favourite / Wishlist Feature	TC-0034	1. Login as customer 2. Open any product 3. Click the heart/favourite icon	Product: Any listed item	Product should be added to favourites and heart icon should turn filled/red	Heart icon shows visual change but on page reload the item is not in favourites; state not persisted	Medium	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Favourites saved to local state only; no API call to persist
25	BG-0026	Admin Panel	Unauthorized Admin Access	TC-0035	1. Login as regular customer 2. Manually navigate to /admin/dashboard in URL bar	User role: customer (non-admin)	Access should be denied; user should be redirected to homepage or shown 403 Forbidden	Admin dashboard loads successfully for a non-admin customer; full admin UI accessible	Critical	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Role-based access control not enforced on admin route
26	BG-0027	User Login	Forgot Password Flow	TC-0036	1. Navigate to /auth/forgot-password 2. Enter registered email 3. Click Send Reset Link	Email: customer@practicesoftwaretesting.com	System should send a password reset email and show confirmation: 'Check your inbox'	Confirmation message is displayed but no reset email is received; link is never sent	High	Open	Browser: Chrome 120 OS: Windows 11 Resolution: 1920x1080	QA Team	QA Team	QA Team	[Screenshot]	Password reset email trigger not implemented or mailer misconfigured

Figure 3: Screenshot showing Test Cases TC-0021 to TC-0025 from the Excel Sheet

6 Bug Identification

6.1 Complete Bug Summary Table

Bug ID	Module	Summary	Severity	Status
BG-0003	Navigation	Category filter has no effect on product listing	High	Open
BG-0004	Search	Valid keyword search returns zero results	High	Open
BG-0005	Product Listing	Sort by price has no effect on display order	Medium	Open

Bug ID	Module	Summary	Severity	Status
BG-0006	Product Details	Product image is broken; URL returns 404	Medium	Open
BG-0007	Shopping Cart	Cart header counter does not update after add	High	Open
BG-0008	Shopping Cart	Quantity of zero accepted; item stays in cart	High	Open
BG-0009	Shopping Cart	Cart total calculated incorrectly for qty > 1	Critical	Open
BG-0010	User Registration	Weak 3-character password accepted at register	High	Open
BG-0011	User Registration	Duplicate email registration gives no error	Medium	Open
BG-0012	User Login	Blank page shown after successful login	High	Open
BG-0013	User Account	Profile changes are not saved to the backend	High	Open
BG-0014	User Account	Change password form silently fails	High	Open
BG-0015	Contact Form	Empty form submits successfully without validation	High	Open
BG-0016	Contact Form	Invalid email format accepted on contact form	Medium	Open
BG-0017	Checkout	Unauthenticated guest can access checkout	Critical	Open
BG-0018	Checkout	Blank payment details accepted; order placed	Critical	Open
BG-0019	Checkout	No order confirmation email sent after checkout	Medium	Open

Bug ID	Module	Summary	Severity	Status
BG-0020	Product Details	Out-of-stock items can still be added to cart	High	Open
BG-0021	Product Listing	Pagination shows duplicate products on page 2	Medium	Open
BG-0022	Product Details	Submitted reviews not displayed after refresh	Medium	Open
BG-0023	User Account	Order history is blank for users with past orders	High	Open
BG-0024	Search	XSS vulnerability via unsanitised search input	Critical	Open
BG-0025	Navigation	Wishlist items not persisted after page reload	Medium	Open
BG-0026	Admin Panel	Admin dashboard accessible by regular customers	Critical	Open
BG-0027	User Login	Forgot password reset email is never sent	High	Open

Table 1: Complete Bug Identification Table — BG-0003 to BG-0027

6.2 Severity Distribution

Severity	Bug Count	Share
Critical	5	20%
High	10	40%
Medium	10	40%
Total	**25**	**100%**

6.3 Defects by Module

Module	Bugs Found
User Registration & Login	4
Product Listing, Search & Filtering	4
Shopping Cart	3
Product Details	3
Checkout & Payment	3
User Account (Profile / Password / Orders)	3
Contact Form	2
Navigation & Wishlist	2
Admin Panel	1
Total	**25**

6.4 Critical Bugs: Immediate Action Required

The five bugs classified as **Critical** must be resolved before any production release. They are described below:

1. **BG-0009 — Incorrect Cart Total Calculation**
   A pricing logic error causes the cart to compute wrong totals when a product has a quantity greater than one. Customers could be shown and charged an incorrect amount, which has direct financial and legal implications.

2. **BG-0017 — Checkout Accessible Without Authentication**
   The checkout route has no authentication guard. An anonymous guest user can navigate directly to the checkout page, fill in the form, and potentially submit an order without any user account, bypassing all account-linked controls.

3. **BG-0018 — Blank Payment Details Accepted at Checkout**
   The checkout form imposes no validation on payment fields. Submitting the form

with every payment field empty still results in a successful order confirmation, which is a critical business logic and financial processing failure.

4. **BG-0024 — XSS Vulnerability via Search Input**
   The search bar does not sanitise user-supplied input before rendering it to the DOM. A script tag injected into the search field executes directly in the browser, exposing every user to potential Cross-Site Scripting attacks, session hijacking, and data theft.

5. **BG-0026 — Admin Dashboard Accessible by Regular Customers**
   Any authenticated customer can navigate directly to `/admin/dashboard` and gain full administrative access. There is no role-based access control check on this route, representing a total failure of the authorisation layer.

7 Conclusion

This laboratory experiment successfully demonstrated the end-to-end process of structured manual black-box testing on a practice e-commerce web application. A total of **25 defects** were discovered, documented, and classified across **9 functional modules** of the Toolshop v5.0 (buggy) application, comprising 5 Critical, 10 High, and 10 Medium severity issues.

Key findings from this experiment:

- **Input validation is the most pervasive failure.** Multiple forms across the application — registration, login, contact, cart, and checkout — accept empty, malformed, or malicious inputs without triggering any client-side or server-side validation. This single category of defect accounts for the majority of bugs found.

- **Critical security vulnerabilities are present.** The XSS vulnerability in the search bar (BG-0024) and the fully unprotected admin route (BG-0026) demonstrate that security was not considered during development. These require immediate remediation and cannot be deferred.

- **Core e-commerce workflows are non-functional.** Searching products, sorting by price, cart total calculation, and pagination are all broken. These are not edge-case scenarios but the primary user journeys of any online store, making the application unusable in its current state.

- **Data persistence is unreliable throughout.** Profile updates, password changes, product reviews, wishlist items, and order history all exhibit a recurring pattern of frontend state not being synchronised with the backend API. This points to widespread missing API calls or unhandled response errors.
- **Authentication and authorisation are insufficiently implemented.** The absence of an auth guard on the checkout route, the open admin panel, and the broken post-login redirect collectively indicate that the security architecture was either not implemented or not tested.

Lessons reinforced by this experiment:

(a) Designing test cases before execution ensures full module coverage and prevents ad-hoc, incomplete testing.

(b) Classifying bugs by severity immediately focuses developer effort on the defects that carry the highest risk to users and the business.

(c) Security test cases — input injection, access control checks, unauthenticated route access — must be part of every standard functional test cycle, not treated as a separate activity.

(d) Every data operation must be verified at both the UI level and at the API level. A passing UI message does not confirm a successful backend write.

All 25 reported bugs are in **Open** status and are documented in the accompanying Excel bug report (BG-0003 through BG-0027) with full reproduction steps, test data, expected and actual results, test environment details, and screenshot placeholders for evidence attachment.